

CARE-DN – Explanation & Elaboration (E&E)

Purpose of this document

This Explanation & Elaboration (E&E) document accompanies the CARE-DN checklist. It provides item-by-item guidance and illustrative examples to help authors apply each dry-needling (DN)–specific item consistently when reporting case reports. It should be read alongside:

- the original CARE checklist, and
- the CARE-DN extension, which adds 16 DN-specific items.

The E&E explains why each DN-specific item matters, what should be reported, and offers short examples or wording templates. Examples are illustrative, not prescriptive. Authors should adapt them to their own clinical context and journal requirements.

“Dry needling” as umbrella term

Always use *dry needling* as the core term. This ensures clarity, indexing and consistency across reports. Where helpful, a single modifier may be added (for example *trigger point dry needling* or *intramuscular dry needling*) to clarify the primary anatomical target or technique, provided the terminology is used consistently in the title, abstract, keywords and Methods.

Hygiene and anatomical considerations (contextual reporting)

CARE-DN is a reporting guideline, not a practice standard. It does not prescribe how DN should be performed or which infection-control procedures must be used. However, authors may report contextual factors that influence safety or interpretation, for example:

- skin preparation or sterile/clean field use,
- special precautions in deep or anatomically complex regions,
- factors relevant to adverse events.

Such details are optional and should be included when they clarify the clinical context, particularly in relation to item 10d of the original CARE guideline (adverse and unanticipated events), not as separate checklist items.

TITLE

Item 1b – Include “dry needling” in the title

Explanation

Including *dry needling* in the title makes the intervention explicit, improves indexing and helps readers and reviewers recognise DN case reports quickly. This directly supports discoverability and linkage with DN literature and MeSH indexing.

Example

“Resolution of chronic plantar heel pain after dry needling: A case report”

If a single modifier is central to the case, authors may optionally use expressions such as *trigger point dry needling* or *perineural dry needling*.

KEYWORDS

Item 2b – Include “dry needling” in the keywords

Explanation

Standardised keywords enhance searchability and facilitate identification of DN-related case reports. Including *dry needling* as a keyword aligns with indexing conventions and helps systematic reviewers and database searches. *Dry Needling* is also an official MeSH descriptor (UI: D000079245), so its inclusion supports accurate indexing in biomedical databases.

Example

Keywords: dry needling, myofascial pain, shoulder, case report

ABSTRACT

Item 3e – Specify the DN intervention clearly in the abstract

Explanation

The abstract is often the only part many readers see. Explicitly naming DN and briefly specifying the type of DN intervention improves clarity and discoverability. Therefore, within tight word limits, DN should be named clearly and, when central to interpretation, a concise target or technique modifier may be used.

Practical guidance

- Always name *dry needling* in the abstract.
- Where essential, add one short modifier (for example *intramuscular dry needling*, *intratendinous dry needling*, *ultrasound-guided dry needling*), ensuring terminology matches the detailed report in items 9d–9m.

Example

“Treatment consisted of ultrasound-guided intratendinous dry needling and exercise therapy.”

TIMELINE

Item 7b – Report the number of DN sessions delivered and the interval between sessions. If intervals changed during care, report what changed and why.

Explanation

Reporting how many DN sessions were delivered and how frequently they occurred provides essential context for treatment dosage and allows readers to compare cases and outcomes. Changes in planned frequency can affect interpretation (for example, early improvement leading to fewer sessions, or interruptions due to illness or travel). This information also

enhances reproducibility, enabling other clinicians and researchers to approximate the treatment dose and schedule used in the case.

What to report

- Total number of DN sessions.
- Planned interval (for example weekly, twice monthly).
- Actual intervals if they changed, plus a brief explanation if this is relevant to interpretation.
- Ensure consistency with the timeline in CARE item 7a.

Example

“Six DN sessions were planned at weekly intervals. The patient attended four sessions over 6 weeks due to an intercurrent respiratory infection and work-related travel.”

DIAGNOSTIC ASSESSMENT

Item 8e – Rationale for DN: clinical reasoning behind selecting DN over other treatments

Explanation

Describing why DN was chosen over alternative options improves transparency, supports evidence-based reasoning and helps readers understand the context in which DN was used.

What to report

- Why DN was considered appropriate at this stage.
- Relevant prior interventions and their results (for example limited response to manual therapy or exercise) – if applicable.
- Any key mechanistic rationale (for example targeting a palpable trigger point or tendinopathic lesion).
- Patient-specific factors influencing the choice (for example comorbidities, contraindications, preferences).

Example template

“We selected dry needling to address [target or impairment] after [relevant prior care or context] because [reason DN was suitable for this case].”

Example

“We selected dry needling to address proximal patellar tendinopathy after limited response to progressive loading, as ultrasound-guided intratendinous needling targeted the degenerative focus and the patient declined injection therapy.”

Item 8g – Report any imaging or objective assessments used (if any) to guide diagnosis or dry needling (e.g. ultrasound, EMG, dynamometry).

Explanation

This item applies only when such tools were used. Its purpose is to make transparent any imaging or objective assessments that informed diagnosis or the decision to use DN, or that were used to monitor response. It does not require the use of imaging or devices.

What to report

- Type of tool (for example B-mode ultrasound, EMG, dynamometer, pressure algometer).
- Purpose (for example confirming diagnosis, identifying target tissue, monitoring strength or pain thresholds).
- For imaging, a brief indication of key technical aspects relevant to interpretation, such as transducer type and frequency, depth range or mode, recognising that detailed device reporting for procedural guidance is captured more specifically under item 9j.

Example

“Diagnostic ultrasound (B-mode, 12 MHz linear probe, depth 3 cm) demonstrated focal hypoechoic thickening of the proximal patellar tendon, which guided the decision to perform intratendinous DN.”

THERAPEUTIC INTERVENTION

Item 9d – Report targeted muscles/tissues (name each muscle treated)

Explanation

Accurate replication requires explicit identification of all muscles or other tissues treated. Broad descriptions such as “shoulder muscles” are insufficient. Anatomical clarity also helps contextualise outcomes and any adverse events.

What to report

- Each muscle or structure treated by name.
- Non-muscular targets (for example tendon, scar tissue, peri-neural tissues) where applicable.

Example

“Dry needling was applied to the upper trapezius, infraspinatus and supraspinatus muscles.”

Item 9e – Report anatomical landmarks used for localization

Explanation

Anatomical landmarks help readers understand how the target was located, particularly in deep or complex regions. This supports reproducibility and allows appraisal of anatomical safety.

What to report

- Key surface or palpable landmarks used to localise the target.

- When relevant, any notable anatomical variations encountered.

Example

“The trigger point in the upper trapezius was localised midway between the C7 spinous process and the lateral third of the clavicle, with the patient seated.”

Item 9f – Report needle specifications (brand, material, gauge, length)**Explanation**

Needle characteristics influence depth, manipulation and patient comfort. Reporting brand and physical characteristics supports replication and interpretation of safety considerations.

Example

“Single-use stainless steel needles (Seirin J-Type, 0.25 × 40 mm) were used for all DN applications.”

Item 9g – Number of needles used**Explanation**

The number of needles used in a session is part of the overall treatment dose and may influence both clinical effect and risk of adverse events. Reporting it contributes to a clearer picture of the intervention, especially when multiple muscles or regions are treated.

Example

“Three needles were inserted in the upper trapezius and two in the infraspinatus during each session (total of five needles per session).”

Item 9h – Report depth and angle when relevant and include patient position and the approach path. Numeric depths are not required.**Explanation**

Depth, angle and approach path are important for understanding both therapeutic intent and anatomical risk, particularly near neurovascular structures, pleura or viscera. Numeric depth values are not mandatory, but enough detail should be provided to aid interpretation and adverse event assessment.

What to report

- Patient position (for example prone, supine, side-lying, seated).
- General depth (for example “superficial to fascia”, “to muscle belly”, “to bone contact”, “approximately one-third of needle length”).
- Direction or approach path (for example medial-to-lateral, caudal-to-cranial).

Example

“With the patient prone, the needle was inserted perpendicular to the skin into the upper trapezius to the depth of the muscle belly, directed slightly caudally, with care to avoid the lung field.”

Item 9i – Report the needling technique employed (e.g., static retention, pistoning, rotation) and the duration of active manipulation and, if recorded, number of bouts or frequency.

Explanation

Different DN techniques (for example fast-in/fast-out pistoning, static retention, rotational techniques) and the duration/frequency of active manipulation may influence outcomes and adverse effects. Authors are asked to report both the technique and the duration of manipulation and, when recorded, the number of bouts or frequency.

What to report

- Technique (for example static retention, fast-in/fast-out pistoning, rotation).
- Duration of active manipulation (for example seconds of pistoning).
- If recorded, number of bouts or repetitions.

Example

“A fast-in/fast-out pistoning technique was used for approximately 20 seconds per point, with two to three bouts per needle, followed by static retention for 2 minutes.”

Item 9j – Report imaging guidance if used and include the modality, brand, model, and key settings.

Explanation

This item applies only when imaging is used to guide the procedural delivery of DN (for example ultrasound-guided peritendinous or intratendinous needling). Transparent reporting of imaging guidance supports reproducibility and safety appraisal. Brand, model and key settings shall be reported.

What to report

- Imaging modality (for example ultrasound, fluoroscopy).
- Device brand and model.
- Key settings that materially affect procedure (for example probe type and frequency, depth, main mode or specific safety-related feature).

Example

“Patellar tendon dry needling was performed under real-time ultrasound guidance (GE Logiq e, 12 MHz linear transducer, depth 3 cm, B-mode, focus at the proximal tendon). The needle path and tip were visualised throughout the procedure.”

This example shows clearly that authors should specify modality, brand, model and key settings.

Item 9k – Report whether a local twitch response was elicited (yes/no) and, if recorded, the approximate number and location.

Explanation

The presence or absence of a local twitch response (LTR) is widely reported in DN practice and may be clinically relevant in some protocols. Reporting whether an LTR occurred, and where, helps readers understand how the technique was applied and may enable future analyses on the relationship between LTR and outcomes.

What to report

- Whether an LTR occurred (yes/no).
- If recorded, approximate number and muscles/regions in which LTRs occurred.

Example

“Two local twitch responses were elicited in the infraspinatus and one in the upper trapezius during the first session. No LTRs were observed in subsequent sessions.”

Item 9l – Duration of needle retention**Explanation**

In protocols using static needle retention, the length of time the needle is left in situ may influence effects and patient tolerance. Reporting retention time contributes to understanding treatment dose.

What to report

- Approximate retention time per needle or per region.
- Whether retention time varied across sessions (if relevant to interpretation).

Example

“After active manipulation, needles were retained in situ for approximately 5 minutes in each treated muscle.”

Item 9m – Clinician credentials/experience**Checklist text**

“Clinician credentials/experience.”

Explanation

Clinician professional background, DN training and experience may influence both outcomes and safety, particularly for complex anatomical regions or imaging-guided procedures. This item does not serve as a credentialing standard but promotes transparency about who delivered the intervention.

What to report

- Professional role (for example physician, physiotherapist).
- DN-specific training or certification (for example accredited course, postgraduate module).
- Approximate years of DN practice or experience.

Example

“The intervention was delivered by a physiotherapist with 10 years of musculoskeletal practice and 7 years of DN experience, including completion of an accredited 50-hour DN training programme.”

Closing note

The CARE-DN E&E is intended to be used alongside the CARE-DN checklist and the original CARE guideline. Together, they support more transparent, complete and reproducible reporting of dry needling case reports and provide a structured foundation for future evidence synthesis and methodological improvement in DN research.